Structural Mechanics: A Fail-Closed Applicability Contract for Experience Transfer—Specification, Typed Refusal, and Instrument Falsifiability

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Abstract

Orion persists task episodes, consolidates scoped lessons and reuses experience in a typed external substrate; the companion epistemic-mechanics paper establishes that recorded experience, routing priority and representation changes do not by themselves carry scientific authority. This paper studies the transfer relation inside that substrate: when may experience recorded in one problem fibre move into another? We make three deliberately scoped contributions and leave the generality question explicitly open. First, a *specification*: a directional applicability contract whose witness binds quantity of interest, role/relation mapping, preserved invariants, explicitly non-preserved properties, target boundaries, evidence and uncertainty, and whose non-compensatory decision separates `LICENSED`, `REJECTED` and `CANNOT_CHECK`. We position the contract against its acknowledged formal parent, causal transportability (Bareinboim and Pearl): directionality, QoI binding, source preconditions, invariance declaration, target demarcation, evidence-bearing output and fail-closed refusal are all already occupied by the *sID* signature, and the surviving residuals are exactly four—a returned third verdict when the contract's own preconditions are unverifiable, cross-vocabulary role correspondence established under a licence rather than assumed, operation over non-causal research artifacts without a selection diagram, and explicit forbidden-loss enumeration. Second, a *measured adapted method transfer*: faithfully importing the parent's completeness-style refusal semantics—treating every rejection as an impossibility certificate—is refuted in the witness setting. On a frozen eight-arm known-world design the faithful import emitted false impossibility certificates on all four repairable, open-world and budget-exhaustion arms, while the adapted mechanism, which splits `REJECTED` into `MERELY_UNLICENSED` and `CERTIFIABLY_IMPOSSIBLE` and derives the strong verdict only from witness-independent obstructions under a closed-world declaration and a completed bounded exhaustion, was correct on every runnable arm, with a 600-case proposition sweep (zero disagreements) and 178 planted-pass plus 178 planted-fail known-answer cases (zero failures). Third, a *methods contribution on instrument falsifiability* whose evidence is our own preserved negative results. A frozen six-family confirmatory extension passed every registered gate, after which an audit battery showed the gate was non-falsifiable: one comparison arm was the gold function itself (per-item loss variance 1.2×10^{-37}), twelve of twelve arbitrary seeds passed, two thirds of the headline gain was constructed rather than measured, and scrambling all task text left the gold decision and every coordinate unchanged on 810 of 810 cases. A controlled-witness instrument scoring exact decision 1.0 was shown to measure serialization fidelity, not extraction: its render/extract composition is the identity on the structural record, byte-identical on 2592 of 2592 surfaces. A successor prose instrument built to the resulting falsifiability battery did read its text (scrambling collapsed exact accuracy from 0.9722 to 0.2500) and rejected all three of its negative controls, yet failed the registered error-diversity gate: all 55 errors fell in the single held-out lexicon class, and a matched-pair control attributed the entire measured error to surface wording—template inversion, not extraction. Every negative result is preserved verbatim, and the battery is distilled into design requirements for transfer benchmarks. No cross-domain transfer or generality claim is made. The external-label requirement itself was then executed once through a battery-compliant instrument: on 1,542 third-party-labelled narrative pairs from a published analogy corpus, with the reducer admitted under an author-independence gate and every battery probe passing, the witness arm's advantage was -0.016 with 95% interval $[-0.0498, +0.0162]$ and valid-transfer retention 0.023 at false-accept 0.044—an honest negative that relocates the open generality question from label availability to a measured extraction-capability gap: cross-domain analogies share almost no surface vocabulary, and no admissible reducer in this programme recovers system-level structure from natural prose. The named successor is a capable learned reducer entering through the same admission gate with contamination accounting.

1 From epistemic mechanics to the transfer relation

The companion paper in this series formalizes the epistemic mechanics of an evidence-governed research substrate [1]. Its results fix the ground this paper stands on: claims are indexed by context, evidence, uncertainty and provenance; scientific authority is a partial order that cannot be faithfully collapsed into a scalar; a projection that maps two worlds requiring different canonical updates to the same represented state admits no correct deterministic policy; and—the invariant this paper inherits directly—recorded experience, retrieval priority and representation changes may alter computational access and search cost without conferring evidential authority. Orion, the system both papers describe, persists task episodes, consolidates scoped lessons and reuses operators across problems as an implemented architectural fact.

Given that substrate, the next mechanical question is the *transfer relation itself*: under what conditions may an episode, lesson or operator recorded in one problem fibre move into another? The tempting answer—semantic similarity—is known to be unsafe: surface vocabulary can be shared where the load-bearing structure differs, and disjoint where it agrees. The saturation results of the companion paper sharpen the demand: once the epistemic view of a research programme reaches bounded saturation, further progress must come from moving validated structure *across* fibres rather than re-deriving it within one, so the transfer relation becomes the binding constraint on reuse. What is needed is a relation with an explicit scope, an explicit refusal mode, and instruments capable of falsifying claims made about it.

This paper reports what has actually been earned on that question, and no more. Three contributions survive our own audits; a fourth question is left explicitly open.

1. **A specification (Section 2).** The applicability contract: a directional structural witness binding quantity of interest (QoI), role/relation mapping, preserved invariants, explicitly non-preserved properties, target boundaries, evidence and uncertainty, evaluated by a non-compensatory three-verdict decision (`LICENSED / REJECTED / CANNOT_CHECK`). This is a specification claim with an exact scope, not an empirical superiority claim. Section 3 places it against its acknowledged formal parent—causal transportability—and narrows the residual novelty to the four conjuncts that survive that comparison.

2. **A measured adapted method transfer (Section 4).** The parent's strongest property is that refusal is a *certificate of impossibility.* We show by construction that importing this property faithfully into the witness setting is wrong—the faithful import manufactures false impossibility certificates on every repairable, open-world and budget-limited arm of a frozen eight-arm design—and that an adapted typed refusal, which separates `MERELY_UNLICENSED` from `CERTIFIABLY_IMPOSSIBLE`, is correct on every runnable arm under hard gates that could have failed and did not.

3. **A methods contribution: instrument falsifiability (Section 5).** Three of our own transfer instruments were audited after passing, or while passing, their registered gates. One

confirmatory study *passed every frozen gate* and was then shown to be non-falsifiable; one perfect score was shown to measure serialization fidelity; one successor instrument, built to the resulting battery, was falsifiable, read its text, rejected its negative controls—and still failed its error-diversity gate, isolating its measured performance to template inversion. We preserve all three negatives verbatim and distill the battery (gold-arm distinctness, seed-failability, sampled discriminating coordinates, text-destruction probes, matched-pair attribution) into design requirements for transfer benchmarks. We regard these negatives, and the battery they validate, as the paper's most reusable result.

4. **An open question, now with its blocker measured (Sections 7 and 8).** Whether the contract's coordinates can be recovered from natural artifacts, and whether witness gating adds value on independently labelled natural-domain cases, is *not established.* The frozen 16-item natural-domain packet is registered as power-limited for the 0.05 paired-Brier minimum detectable effect. The third-party-label route was executed once under a battery-compliant instrument on 1,542 externally labelled narrative pairs (ARN): the admission gate passed, the battery passed, and the outcome is an honest negative—`NEGATIVE__CAPABILITY_ABSENT`—which relocates the residual from label availability to a measured extraction-capability gap (Section 7.4). No result reported here substitutes for a capable, admissible reducer.

Throughout, same-context analysis—including our own audit batteries—is labelled as such and is not represented as independent review. Negative results are never rewritten; successors are new versioned objects.

1.1 Terminology at a glance

For readers from machine learning or systems engineering, the following plain-language glosses fix the vocabulary used throughout; formal definitions appear at first technical use.

Quantity of interest (QoI)
: the specific target quantity a transfer is meant to preserve; correctness is judged relative to it, not to overall similarity.

Directional structural witness
: the typed object that licenses moving structure from a source to a target: a role map, the invariants that must be preserved, the properties that are *not* preserved, target boundary conditions, and evidence.

Obligation set
: the witness expanded into an executable checklist of transfer obligations, each evaluated `SATISFIED` / `VIOLATED` / `UNKNOWN`.

Non-compensatory
: a single violated load-bearing obligation forces rejection; strength elsewhere cannot buy it back—there is no weighted average.

`LICENSED` / `REJECTED` / `CANNOT_CHECK`
: the three fail-closed verdicts: transfer allowed, transfer refused on a violation, or abstention when required evidence is missing (`UNKNOWN` $\neq$ `VIOLATED`).

`MERELY_UNLICENSED` / `CERTIFIABLY_IMPOSSIBLE`
: the typed split of rejection: this witness failed, versus no admissible witness can succeed—the latter only under a closed-world declaration and a completed bounded exhaustion.

Fibre
: the retrieval index/representation scoped to a fixed (QoI, context) into which an episode is moved.

Forbidden loss
a property whose loss under transfer invalidates the reuse even if the mapping otherwise fits.

Demoted evidence
an evidence path deliberately denied confirmatory/independent authority (e.g. an AI operator standing in for an absent independent annotator).

Template inversion
the failure mode in which an instrument's score measures how much of its surface its author templated, because the extractor inverts the renderer's own constructions.

MDE
minimum detectable effect—the smallest effect size a frozen test is powered to detect.

2 The applicability contract

This section states the contract as a specification: a typed object, a decision procedure, and an exact scope. Nothing in this section is an empirical claim.

2.1 A scoped structural transfer object

A reusable structure should not be a bag of similar words. Let a structural object be

$$S = (V, E, \chi, \gamma, q, \varepsilon),$$

where V is a set of typed roles, E a set of typed directed or undirected relations, χ a set of invariants or constraints, γ a boundary/context record, q the quantity of interest, and ε the evidence identities supporting the extraction. Domain-facing entities are retained in provenance, but the transfer layer reasons over roles and relations.

The object is intentionally scoped. A hospital and a packet network need not be globally "the same." They may instantiate the same queue-stability substructure for a backlog or delay QoI while differing radically in ethics, priority policies, intervention constraints and entity semantics. The formalism therefore uses a directional mapping witness rather than a global equivalence relation.

Definition 1 (Directional structural witness)**.** *For source S_A and target S_B, a witness is*

$$w_{A\to B} = (\mu_V, I^+, I^-, B, E_w, U_w),$$

where μ_V maps source roles to target roles, I^+ records preserved invariants, I^- records explicitly non-preserved properties, B contains required target boundary conditions, E_w is evidence for the mapping and U_w records uncertainty. The witness is licensed only for the registered QoI.

The relation is directional because a transformation useful in A may require information or interventions unavailable in B; a valid mapping from source to target need not license the reverse mapping because scope, information loss or boundary conditions can be asymmetric. It is context-specific because a mapping valid in one regime can fail after a boundary change. It need not be transitive: $A \to B$ and $B \to C$ do not automatically license $A \to C$ unless the composed invariant and boundary obligations are separately satisfied. Directionality itself is not a novelty claim—the causal-transportability parent of Section 3 is directional by construction; what the witness adds is that the correspondence between vocabularies is *established* by the mapping object rather than assumed as shared variable identity.

2.2 Executable transport obligations and fail-closed abstention

A directional witness is interpreted as a finite set of typed transport obligations

$$\Omega^{(q)}_{A \to B} = \{o_1, \ldots, o_m\},$$

scoped to quantity of interest q. An obligation records a role, relation, invariant, source precondition, target boundary, QoI requirement or forbidden loss together with its requirement class, evidence identities and a three-valued verification status

$$\mathrm{status}(o_i) \in \{\mathrm{SATISFIED}, \mathrm{VIOLATED}, \mathrm{UNKNOWN}\}.$$

The distinction between `VIOLATED` and `UNKNOWN` is load-bearing. Failure to establish a required precondition is not itself evidence that the precondition is false.

The transfer decision is deliberately non-compensatory. Write $\mathrm{Transfer}(A \to B; q)$ for the directional decision under QoI q. Then:

- LICENSED if every load-bearing obligation is satisfied;
- REJECTED if at least one load-bearing obligation is demonstrably violated;
- CANNOT_CHECK otherwise, whenever at least one load-bearing obligation remains unresolved.

Thus several preserved relations cannot compensate for a violated target boundary, wrong transfer direction or forbidden loss. Conversely, a property declared irrelevant to the registered QoI may be listed as a permissible loss rather than forcing global structural equivalence. Each load-bearing obligation is content-bound to evidence rather than inheriting authority merely because the witness object contains some evidence somewhere. The implementation emits an obligation-level trace and a deterministic content hash so that a confirmatory benchmark can freeze the exact witness contract before outcome access. The decision procedure is shown in Figure 1. Section 4 refines the `REJECTED` verdict into a typed pair; the refinement is the paper's measured contribution and is stated there, not here.

2.3 Why explicit non-preservation matters

Analogies are dangerous when the reusable part and the non-reusable part are stored in one undifferentiated similarity score. The witness therefore records I^-. In the queue example, "arrival competes with service and excess load grows backlog" may transfer while domain-specific priority policy and entity semantics do not. This is more than documentation: a later operator can declare which preserved invariants it depends on, and an intervention touching a non-preserved property can fail closed. We found no counterpart to this explicit forbidden-loss enumeration in any of the literature families searched in Section 3; it is one of the four surviving residuals.

2.4 Where the contract sits in the deployed router

Orion's obstruction–transformation router gives the witness a precise operational location. The router fingerprints the active target obstruction using roles, relations, constraints, failure mechanisms, invariants to preserve, desired transition and forbidden losses, then queries a content-bound memory of recorded source episodes $O_s \xrightarrow{T_s} O'_s$. A semantic or structural ranker may nominate a source, but source retrieval is not yet transfer. For a cross-domain `JUMP`, the source transformation becomes a viable target proposal only when a mapping witness accounts for the load-bearing source structure: source-to-target role mapping, shared relations and constraints, every enabling source precondition, material disanalogies and target-domain validation obligations. An unrepaired source precondition or a conflict with a forbidden target

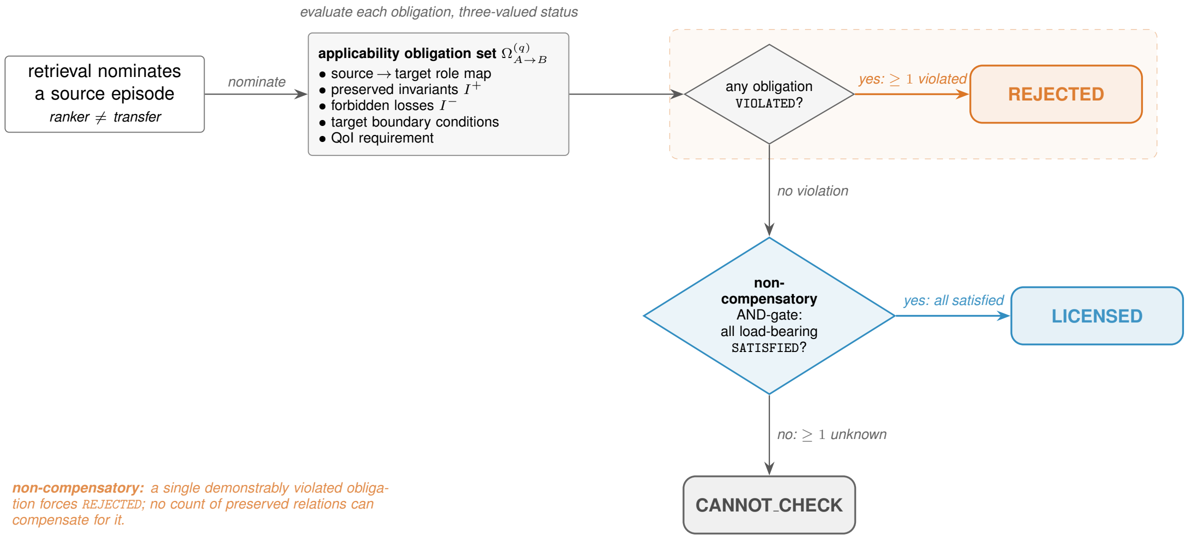

Figure 1: The fail-closed transfer decision. A nominated source is evaluated against the applicability obligation set (role map, preserved invariants I^+, forbidden losses I^-, target boundary, QoI), each with a three-valued status. The rule is non-compensatory: any single demonstrably violated load-bearing obligation forces `REJECTED`; only when every load-bearing obligation is satisfied is the transfer `LICENSED`; an unresolved obligation yields `CANNOT_CHECK`. No count of preserved relations can buy back a violation.

invariant blocks the transfer even when the source event itself is well verified. For `GLUE`, partial episodes whose combined recorded effects cover the desired transition remain composition candidates only; a separate composition witness must bind operation order, interfaces and incompatibility checks. Directional witnesses license transport of components; they do not prove that a composed pipeline is coherent.

This yields the three-stage distinction of Figure 2:

$$\text{retrieved source} \neq \text{witnessed target applicability} \neq \text{verified target success.}$$

This paper studies the middle relation. When a witness passes, Orion may raise retrieval priority, activate an already promoted tool, or place an operator path earlier in the search queue. The permitted inference is witnessed applicability $\Rightarrow$ eligible experience reuse, *not* witnessed applicability $\Rightarrow$ scientific claim true. Scientific authority remains owned by the target problem's verification path—the cross-domain specialization of the companion paper's invariant that experience-conditioned routing is not epistemic authority [1].

2.5 Negative transfer is recorded through an evidence ladder

A failed transfer attempt first creates a non-success task episode. It does not immediately establish that the witness class, operator or source lesson is invalid. The system records the failure as observed-only, then permits competing diagnoses such as context mismatch, broken invariant, implementation defect, missing evidence or an unrelated downstream failure. Only discriminating evidence can promote a supported diagnosis, and only fresh replay/transfer can turn that diagnosis into a reusable boundary lesson. This prevents one near-miss from becoming a global blacklist while still allowing repeated boundary-sensitive failures to improve later routing, and it makes negative-transfer history addressable: a future witness can explicitly inherit a known non-preserved property or target boundary rather than merely receiving a lower opaque score.

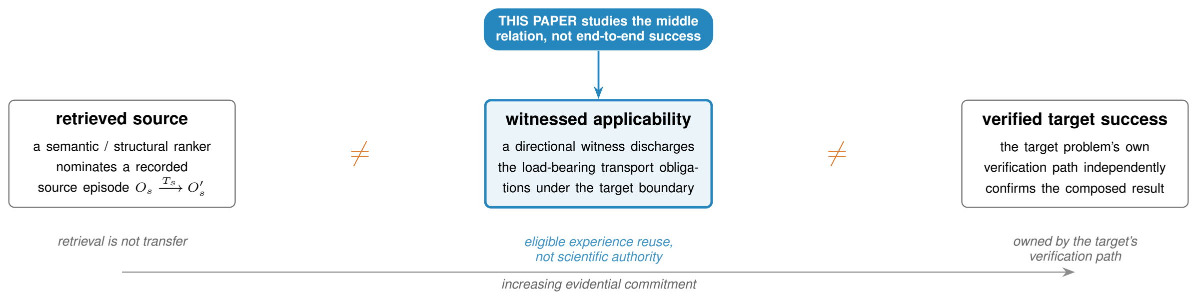

Figure 2: Three stages of increasing evidential commitment. A retriever nominating a source episode is not the same as a directional witness discharging the load-bearing transport obligations under the target boundary, which is in turn not the same as the target problem's own verification path independently confirming success. This paper's subject is the middle relation—witnessed applicability—which licenses eligible reuse without minting scientific authority; retrieval and end-to-end success are owned by other stages.

2.6 Exact scope of the specification claim

The claim of this section is that the contract is (i) executable—the reference implementation evaluates obligation sets deterministically and emits content-hashed traces; (ii) fail-closed—missing evidence yields abstention, not a score; and (iii) exactly scoped—each verdict names the QoI, boundary and obligation set it is conditional on. The claim is *not* that the contract improves transfer decisions on any natural distribution of problems, that its coordinates can be recovered from natural prose, or that its verdicts correlate with downstream research utility. Those are empirical questions; their current evidential status—including the refutation of two of our own instruments and the template-inversion terminal of a third—is reported in Sections 5–7.

3 Parents, the acknowledged formal parent, and the surviving residual

Earlier drafts of this manuscript presented the contract's conjunction—directional, QoI-bound, precondition-accounting, invariant-preserving, boundary-demarcated, evidence-bearing, fail-closed—as residual novelty. A systematic nearest-work audit showed that this conjunction will not survive review, because nearly every conjunct is occupied by one parent formalism. This section records that comparison explicitly and narrows the claim to what survives. The audit artifacts, including primary-source verification for every entry, are preserved in the repository (`research/paper2_nearest_work_2026/`).

3.1 Causal transportability is the acknowledged formal parent

The transportability theory of Bareinboim and Pearl [2, 3, 6, 7] asks when a causal effect learned in a source population Π can be transferred to a target population Π^*, given a selection diagram D demarcating where the two populations may differ. The *sID* algorithm decides this question, is provably complete for its class [3, 5], extends to multiple source domains [4], and on failure returns a structural object witnessing the obstruction. Partial transportability yields bounds when point transport fails [8]. Pearl and Bareinboim describe transportability as a licence to transfer causal effects learned in experimental studies to a new population—the same word this paper uses.

Table 1 records the conjunct-by-conjunct comparison. Read plainly: directionality, QoI binding, source-precondition accounting, invariance declaration, target demarcation, evidence-

Contract conjunct	Transportability counterpart	Status
directional	$\Pi \to \Pi^*$, asymmetric by construction	occupied
binds QoI	$P^*_x(y)$ is an explicit formal argument	occupied
source preconditions	the available-distribution inputs $\langle P, I, P^* \rangle$	occupied
preserved invariants	absence of an S-node *is* declared mechanism invariance	occupied
target boundaries	the selection diagram D demarcates what differs	occupied
evidence-bearing	returns a transport formula naming which datasets to fuse	occupied
fail-closed	FAIL with an s-hedge: refusal is a proof of impossibility	occupied, strictly stronger
not similarity-based	do-calculus derivation, zero similarity scoring	occupied
uncertainty at target	partial transportability yields bounds	largely occupied

Table 1: Conjunct-by-conjunct comparison of the applicability contract against the *sID* signature of causal transportability. Every conjunct of the earlier residual-novelty sentence is occupied by the parent; the surviving residuals are the four listed in Section 3.3.

bearing output and fail-closed refusal are all already present in the *sID* signature, and on refusal the parent is strictly *stronger* than fail-closed—completeness turns refusal into a certificate of impossibility, whereas a fail-closed gate merely declines. Any causal-inference reviewer would read the earlier residual sentence as the transport formula with different nouns. We therefore cite transportability as the acknowledged formal parent of the applicability contract, not as related work.

3.2 Neighbouring families occupying single conjuncts

Structure mapping. Structure-Mapping Theory and its computational lineage—SME and MAC/FAC [9, 10, 11]—own the role/relation-mapping conjunct outright, with systematicity as a principled invariant-selection rule. The output differs: SME produces ranked candidate mappings, not a licence. The contract may not present role mapping itself as novel, and does not.

Selective prediction and deferral. Learning with rejection, selective classification, learning-to-defer and conformal prediction under shift [12, 13, 14, 15] own the abstain channel, including abstention mounted directly on transfer settings with target-domain coverage guarantees. The trigger differs: abstention there is threshold-triggered on a scalar confidence signal, whereas `CANNOT_CHECK` is structure-triggered by a named unverifiable obligation. That difference is a claim only insofar as it is tested against a matched-coverage selective-prediction baseline (Section 8).

Case-based reasoning. Adaptability-guided retrieval [17], within the classical CBR frame [16], is the ancestor of precondition-bound reuse: it replaces semantic similarity with estimated adaptability as the retrieval criterion. It produces a real-valued estimate feeding ranked retrieval, not a directional contract, and has no abstention verdict.

Contemporary reusable-experience governance. SKILL.nb [18] is the nearest live contemporary: evidence-calibrated lifecycle policies for reusable agent workflows, gate-conditioned execution and fallback on drift—the same problem statement and the same fail-closed instinct. It differs in the decision object: binary runtime step validation, no declared QoI, no role mapping, no forbidden-loss enumeration and no third verdict. More broadly, shared experience substrates and dependency-aware skill retrieval are established parents rather than contributions of this paper: ReX supplies reusable latent experiences composed into lightweight adapters [19]; SkillGraph-style systems supply dependency, prerequisite and precondition–effect graphs that inform retrieval, composition and execution [20, 21, 22, 23]; and SkillSight shows that shared descriptive boilerplate inflates semantic relevance scores, which is why any semantic control used against the contract must be calibrated against shared-background bias [24].

Analogy evaluation. The analogy/transfer benchmark literature [25, 26, 27, 28, 31, 30, 29] measures whether transfer succeeds or ranks candidate analogies by similarity; AbstentionBench [32] abstains on answerability rather than on transfer. None of these issues a per-transfer licence, and none is a threat to the narrowed claim; they are cited as the evaluation context the open programme must engage. Empirically, controlled asymmetry in structural transfer between natural-language and biological foundation models [33] is a direct counterexample to the assumption that shared structural regularity implies symmetric representational transfer, which is why the witness defaults to a directed edge and why direction reversal is a mandatory perturbation in every instrument of Section 5.

3.3 The four surviving residuals

After the comparison above, the contract's residual claim is exactly the following four conjuncts, and nothing broader.

1. **A returned third verdict.** Transportability is complete *conditional on a fully specified selection diagram*; it has no output value meaning "I cannot evaluate this." The data-fusion synthesis states plainly that if knowledge about commonalities and disparities is unavailable, transport cannot be justified—and discharges that case to the user, outside the formalism [7]. A contract that *returns* `CANNOT_CHECK` when its own preconditions are unverifiable is outside the parent formalism.

2. **Cross-vocabulary role/relation mapping under a licence.** Transportability presumes shared variable identity—the selection diagram overlays two causal diagrams over the same variable set. SME maps across vocabularies but does not license. The contract must *establish* the correspondence transportability assumes for free, then license across it. No parent found does both; this is the strongest residual.

3. **Applicability without a causal graph, over non-causal research artifacts.** D is a hard input to *sID*, and its QoI is a probabilistic causal quantity. Reusable research experience—episodes, lessons, operators—is neither.

4. **Explicit forbidden-loss enumeration.** A first-class record of properties whose loss invalidates reuse, checked non-compensatorily. No counterpart was found in any of the five searched families.

Each residual is a specification-level claim about what the contract expresses, not an empirical superiority claim. The measured contribution built directly on this comparison—absorbing the parent's refusal semantics by adaptation rather than by faithful import—is Section 4.

4 Typed refusal: absorbing the parent's strongest property by adaptation

The strongest property of the transportability parent is what its refusal *means*. When *sID* fails, completeness upgrades the failure from "this derivation did not succeed" to "no derivation over the declared inputs can succeed," and the returned obstruction witnesses that impossibility [3, 5]. This section reports a measured method transfer of that property into the witness setting: a frozen source-mechanism analysis, a faithful-import control built to fail, an adapted mechanism built to the source's actual preconditions, and a known-world execution in which the contrast between them is the experiment. All artifacts are preserved under `research/paper2_causal_transport_absorption_v1/`; the adapted mechanism is implemented in `src/rakl/transfer_impossibility.py`.

4.1 What makes the parent's refusal a certificate

The source analysis, frozen before implementation, identifies three load-bearing preconditions behind refusal-as-impossibility. First, a *complete declaration* of the objects the verdict ranges over: *sID* requires the selection diagram as a hard input, and the authors place the case of unavailable knowledge outside the formalism entirely. Second, a *complete underlying calculus*: refusal upgrades from "search failed" to "nothing exists" only because do-calculus plus standard probability manipulations are complete for the class. Third, a *decidable exhaustion* of the derivation space, so that "no derivation succeeded" is established rather than assumed. The impossibility itself is proved by an indistinguishability construction—two models agreeing on every declared input yet disagreeing on the QoI—so the certificate derives from properties invariant under every admissible presentation of the query, not from the failure of one presentation. The claimed benefit is correspondingly scoped: non-existence is relative to the fixed input tuple, not absolute; enlarging the declared inputs can flip a refusal into a transport formula.

4.2 The gap in the witness gate, and the adapted mechanism

Orion's applicability gate evaluates exactly one supplied witness. Its `REJECTED` therefore quantifies over nothing: it reports that *this* witness failed and cannot distinguish "no witness can license this transfer" from "this particular witness is defective." Importing the parent's verdict form without its preconditions would manufacture false impossibility certificates—relabelling a refusal caused by a repairable defect in the supplied presentation as proof that no presentation can work. That is precisely the surface-analogy forcing this research programme forbids.

The adapted mechanism splits the negative verdict:

$$\texttt{REJECTED} \longrightarrow \{\texttt{MERELY\_UNLICENSED}, \texttt{CERTIFIABLY\_IMPOSSIBLE}\}.$$

`CERTIFIABLY_IMPOSSIBLE` may be emitted only when (i) the target carries an explicit closed-world completeness declaration—the honest analogue of the fully specified selection diagram; (ii) a bounded exhaustion over the witness presentation space actually completes; and (iii) the obstruction is witness-independent (a QoI mismatch, a declared invariant absent from the target, or a relation type no admissible role mapping can preserve). Budget exhaustion, an open-world target, or any witness-local defect yields `MERELY_UNLICENSED`. The strong verdict carries a certificate naming the failed criterion, the unmatchable relation types or missing invariants, and any role-cardinality obstruction. The cost asymmetry is inherited honestly: quantifying over role mappings is exponential in the worst case, which is why an unfinished exhaustion must produce the weak verdict, never the strong one.

4.3 Frozen falsifiers and the eight-arm execution

Three falsifiers were frozen with the candidate, any one of which kills the transfer: (a) the adapted gate emits a single false impossibility certificate on a case where a licensing witness demonstrably exists; (b) it never fires `CERTIFIABLY_IMPOSSIBLE` on cases genuinely impossible under a closed-world declaration, making it vacuous; (c) it never differs from the faithful import, making the distinction empty. The faithful import (challenger A) relabels every `REJECTED` as `CERTIFIABLY_IMPOSSIBLE` with assumptions unchecked; the adapted gate is challenger B.

Table 2 summarizes the executed receipt. The faithful import produced false impossibility certificates on all four repairable/open-world/budget arms—the registered refutation of the import. The adapted gate emitted no false certificate (falsifier (a) did not fire), fired the strong

Arm	Ground truth	A	B	B certificate
1 repairable role map	`MERELY_UNLICENSED`	false cert.	correct	—
2 repairable evidence	(constructor rejects)	`CANNOT_CONSTRUCT`: evidence-free witness inadmissible		
3 repairable boundary	`MERELY_UNLICENSED`	false cert.	correct	—
4 structural relation	`CERTIFIABLY_IMPOSSIBLE`	correct	correct	S3: unmatchable `causes`
5 structural QoI	`CERTIFIABLY_IMPOSSIBLE`	correct	correct	S1: QoI mismatch
6 structural invariant	`CERTIFIABLY_IMPOSSIBLE`	correct	correct	S2: missing `energy`
7 open-world target	`MERELY_UNLICENSED`	false cert.	correct	—
8 budget exhaustion	`MERELY_UNLICENSED`	false cert.	correct	—

Table 2: The eight frozen arms. The faithful import (A) emits a false impossibility certificate on every repairable, open-world and budget-limited arm; the adapted gate (B) is correct on every runnable arm and emits certificates only on the three witness-independent structural obstructions. Arm 2 is preserved as `CANNOT_CONSTRUCT`: the public constructor rejects an evidence-free witness, so that branch is unreachable and is reported as unreachable rather than counted as a pass.

verdict on exactly the three structural arms (falsifier (b) did not fire), and differed from the import on four arms (falsifier (c) did not fire). Beyond the constructed arms, two sweeps executed under hard gates: a 600-case randomized proposition sweep checking the adapted gate's verdicts against an oracle criterion, with zero disagreements, of which 288 cases carried source relations and 18 discriminated via a non-canonical role mapping; and a planted known-answer sweep of 178 planted-pass and 178 planted-fail cases with zero failures (seed 20260814). All five hard gates passed: no false certificate, non-vacuity, separation, proposition agreement, and planted known-answer agreement.

4.4 What this does and does not establish

Establishes. In the known-world setting, refusal semantics cannot be imported from the completeness parent by surface analogy—the faithful import is refuted by construction on every arm where the refusal cause is witness-local—and the adapted typed refusal, which re-derives the strong verdict from the parent's actual preconditions, is exact on the registered design. This absorbs the parent's strongest property at the level of its quantification structure, never at the level of its causal machinery: Orion has no causal diagram and its QoI is not a probabilistic causal quantity, which is precisely why the adaptation, not the import, is the transferable object.

Does not establish. These are constructed known-answer worlds evaluated in the same research context that built the mechanism; the execution grants no scientific or promotion authority, and no independent reproduction of the transfer exists. The source theorem is peer-reviewed; the transfer is not. Nothing here bears on natural-domain recovery of witnesses (Section 7), and the exponential presentation-space cost means the bounded-exhaustion path must be measured, not assumed, in any deployment claim.

5 Instrument falsifiability: a battery, and three refutations of our own instruments

A confirmatory gate that passes is informative only if it could have failed. This section develops that requirement into an executable battery for transfer instruments and reports its application to three instruments of this research programme—in chronological order of discovery, with every negative preserved verbatim. The first instrument *passed every registered gate* and was then shown to be non-falsifiable. The second scored a perfect 1.0 that turned out to measure serialization fidelity. The third was built to the battery, was demonstrably falsifiable, read its text, rejected its own negative controls—and still failed its error-diversity gate, isolating what it measures to template inversion. We consider the battery, validated by these refutations, the paper's most reusable methodological result.

5.1 The battery

For a benchmark with a gold decision function, candidate arms and a registered paired statistic, the battery comprises:

A. Gold-arm distinctness
No comparison arm may *be* the gold function. If it is, the "paired" statistic is one arm's absolute loss minus a constant, and one arm has no variance.

B. Seed-failability
Re-run the frozen summary on arbitrary non-registered seeds. A registered gate that no seed can fail is a property of the generator, not a test; a probative registration must exhibit at least one failing seed.

C. Trivial-arm floor
Run `always_accept`, `always_reject`, `always_abstain`. Any figure a trivial arm attains (e.g. zero false-accepts from `always_reject`) is worthless alone; only joint properties no trivial arm attains are reportable. Corollary: report false-accept and valid-transfer retention as a pair, always.

D. Equal-n shuffle null
Destroy only the task $\leftrightarrow$ coordinate binding while preserving marginals and the merge rule; the observed arm must lie outside the null. This rules out degenerate generators; it does not establish an extraction or transfer reading.

E. Sole-discriminator map
Leave-one-coordinate-out over the contract. If some coordinate is never the sole discriminator in any stratum, the ablation describes the generator, not the contract.

F. Construction-vs-measurement decomposition
Attribute the headline gain by stratum. A control arm scoring *exactly* 0.000 on a stratum is the signature of strata built so the control cannot see the discriminating coordinate: that share of the gain is constructed, not measured.

G. Text destruction
Replace every task surface with noise and recompute everything. If gold and coordinates are unchanged, the benchmark contains no extraction problem and no arm is solving one. For any extraction reading, destroying the text must destroy performance.

H. Matched-pair surface attribution
Render the same latent draw through two disjoint surface banks. With structure held fixed, any score difference is attributable to surface wording; this decomposes measured error into structural difficulty versus lexicon coverage.

Probes A–G were discovered by auditing the first two instruments below; the third instrument registered A–G analogues (variance, seed-spread, negative-control gates) *before* outcome access, plus H as an auxiliary control. All battery executions are same-context analysis and are labelled as such.

5.2 Case 1: the six-family gate passed—and was non-falsifiable

A six-family robustness extension of the objective known-world lane (Section 6) was frozen as a separate confirmatory coordinate and executed exactly once: seed `2026081212`, $n = 810$ (360 accept / 360 reject / 90 `CANNOT_CHECK`), with freeze integrity verified by blob hash on the execution host before execution. Every registered gate passed with empty gate reasons: positive family residuals 6/6; exact two-sided sign test $p = 0.03125$; paired binary-Brier gain 0.360 with item-bootstrap 95% interval $[0.3267, 0.3933]$ against the registered 0.05 MDE; no retention harm and no false-accept harm in any family, with the full arm retaining every valid transfer at zero invalid false-accepts.

The audit battery then showed that this pass carries no generalization content (`research/paper2_six_family_audit_v1/`):

- **Probe A failed.** The confirmatory runner binds the full arm to the verifier itself: the full arm *is* the gold function, with binary probabilities fixed at 0.98/0.02, so its per-item Brier loss is a constant 0.0004—measured variance 1.24×10^{-37}. The registered "paired" gain is the mechanism arm's absolute loss minus a constant.
- **Probe B failed.** Twelve of twelve arbitrary non-registered seeds reproduce 6/6 positive families, $p = 0.03125$ and a supported verdict. The registered p-value is a property of the generator's item strata; a gate no seed can fail is not a test.
- **Probe F: two thirds of the gain is construction.** The two strata whose discriminating coordinate is by design invisible to the mechanism-only control score mechanism exact-accuracy *exactly* 0.000 and together supply 66.7% of the headline advantage; family-level gains are quantized at exactly 0.48 and 0.24.
- **Probe G: the benchmark contains no extraction problem.** Replacing every source and target text with random noise leaves the gold decision and all six coordinates unchanged on 810/810 cases (only a lexical diagnostic changes, 762/810). The coordinates are read from pre-parsed public fields—string equality, integer comparison, a stored boolean, a float threshold. No arm except the lexical diagnostic reads the text at all.
- **Probes C–D bound what remains.** A trivial `always_reject` arm also attains zero false-accepts, so zero false-accept alone is worthless; the defensible quantity is the joint property (full retention at zero false-accept with correct abstention), which no trivial arm attains. The equal-n shuffle null (200 reps) confirms the generator produces varying task-specific labels—and, given probe G, nothing more.

The honest verdict, preserved in the audit artifact, is SCOPED, not broad generalization: the executed extension demonstrates that the frozen generator, gate and serialization pipeline run to completion and reproduce their registered outputs. It does not support a cross-family transfer law, and given probe G the finding "which applicability coordinates a fail-closed gate must enforce" reduces to arithmetic—checking one of six independent conditions misses violations of the other five.

5.3 Case 2: a perfect score that measured serialization fidelity

A controlled-witness extraction instrument rendered structural tasks into templated text and measured whether an extractor could recover the applicability coordinates: $n = 1296$ base tasks, 2592 text surfaces, gold 576/576/144. The full controlled extractor scored exact decision 1.0, invalid false-accept 0.0, abstention recall 1.0; the strongest non-extraction parent scored 0.889; a ten-mutation panel (dropped fields, span-hash tampering, unknown-state coercions) was caught in full (`research/paper2_controlled_witness_extraction_v1/`).

Probe G refuted the extraction reading. The renderer emits `Label :: serialized-value` lines carrying the structural records verbatim, and the extractor parses them back: the render/extract composition is the *identity* on the structural record, recovered byte-identically on 2592/2592 surfaces, so exact decision 1.0 is entailed by parse success. Scrambling the prose changes every rendered surface (1296/1296) and changes nothing else: gold unchanged 1296/1296, full arm unchanged 2592/2592. The conditional promotion terminal was refuted as an extraction claim; what survives is deliberately narrower and is preserved as such—the mutation panel is a valid test of *fail-closed parser behaviour* under field omission and tampering, an honest terminal of serialization-interface fail-closure, not witness extraction from prose.

5.4 Case 3: a falsifiable instrument that failed honestly—template inversion

The shared mechanism of cases 1 and 2 is that the answer travelled alongside the text in a pre-parsed sibling field, so no arm ever read the text. The successor prose-transfer instrument was therefore registered, *before outcome access*, to the repair specification: an instrument is probative for extraction only if destroying the text destroys performance. Its protocol was frozen as its own commit before any instrument, extractor or runner code existed; gold lives in a latent specification no arm receives and is defined by what the text says—a hedged or elided coordinate is gold-`CANNOT_CHECK`, because that is what a perfect reader of that text would conclude. Seven gates were registered: a lower-bound advantage gate (G1), a strict ceiling < 1.00 (G2), an error-diversity requirement of at least three registered ambiguity classes (G3), a joint retention/false-accept property no trivial arm attains (G4), paired variance in both arms (G5, closing probe A), a twelve-seed spread requirement (G6, closing probe B), and mandatory failure of all three negative controls—scrambled text, shuffled gold, trivial arms (G7). The probe-F repair also held by construction: the violated coordinate is sampled rather than assigned per stratum, and each of the six coordinates appears as sole discriminator (qoi 54, relation 52, effect 52, boundary 44, direction 43, precondition 43 items), so no arm is zeroed by construction.

Executed once on the heldout confirmatory set ($n = 576$): the instrument *reads its text*—scrambling changes every surface and collapses full-arm exact accuracy from 0.9722 to 0.2500. The collapse landed *below* the registered shuffle-null interval $[0.3385, 0.4149]$ rather than inside it, because on noise the extractor fails closed to abstention on every item; this deviation from the registered NC1 wording is recorded as a deviation, not smoothed over—the operative requirement, that G1 fail under scrambled text, held. G1, G2, G4, G5, G6 and G7 passed (advantage 0.4722, McNemar $p = 1.8 \times 10^{-80}$, bootstrap $[0.4323, 0.5156]$; twelve seeds, nine distinct advantage values in $[0.4653, 0.4809]$; all three negative controls failed the gate as required). One honest qualification: the registered strongest non-extraction parent resolved to a constant arm (exact 0.500), so the frozen advantage is measured against a trivial baseline; against the strongest text-reading parent (keyword polarity, exact 0.4601) the advantage is 0.5122—both numbers are stated because the gate as frozen picked the trivial one.

G3 failed, and that is the result. All 55 coordinate errors fall in exactly one of seven registered ambiguity classes—the qualitative-degree class whose confirmatory lexicon was deliberately held out from development (53 relation errors, 2 effect errors); the other six classes show zero errors across 3,077 realizations. Outside the held-out degree lexicon, the extractor inverts the renderer perfectly. The auxiliary matched-pair control (probe H) sharpens the attribution: rendering the same seeds through the development and heldout surface banks holds the latent draw fixed (192/192 identical specifications, 192/192 identical gold) while changing every rendered text (192/192); the extractor scores 1.0000 on the development surface and 0.9792 on the heldout surface. The entire measured error is attributable to surface lexicon, not structural difficulty. The frozen terminal is preserved verbatim: `INSTRUMENT_NOT_PROBATIVE_` `_TEMPLATE_INVERSION`—what the instrument currently measures is how much of the surface its author templated. The 0.9722 is therefore not a result about extraction, and G1's large, highly significant advantage does not rescue it; that is why the gate was registered two-sided. The gate discriminated. It just did not pass. (`research/paper2_prose_transfer_v1/`.)

The registered revival specification is part of the negative record: a successor's renderer must not be authored by whoever writes the extractor (the single-author coupling produced the zero-error classes); every ambiguity class needs a held-out realization, not just one; and the ceiling must not be raised—a successor reporting a *lower* exact score with errors spread across classes is strictly better evidence. Explicitly rejected as a repair: widening the extractor's cue lists, which would move the score toward 1.00 and the error count toward zero, i.e. further into template inversion while looking like an improvement.

5.5 Design requirements distilled

For any transfer benchmark whose gate is meant to be probative, the three cases above force the following requirements, each traceable to a measured failure: no comparison arm may be the gold function (case 1, probe A); the registered gate must be failable, demonstrated by exhibiting a failing seed (case 1, probe B); discriminating coordinates must be sampled, not assigned per stratum (case 1, probe F; repaired in case 3); the task surface must require recovery, not lookup—destroying it must destroy performance (cases 1–2, probe G; achieved in case 3); each coordinate must be the sole discriminator in some stratum (case 1, probe E; achieved in case 3); false-accept must always be reported paired with valid-transfer retention (case 1, probe C); and error attribution must separate structural difficulty from surface coverage via matched pairs and class-diverse held-out realizations (case 3, probes G3/H). An instrument can satisfy all of the falsifiability requirements and still fail its content gate honestly—case 3—which is exactly what the battery is for.

6 The known-world packets, re-read under the battery

The objective known-world lane predates the battery of Section 5. This section preserves its registered outcomes exactly as executed and then states what they are worth under the battery's findings. Nothing here is softened; the numbers stand, and their reading changes.

6.1 The four-family confirmatory packet: registered outcome

The lane was designed so that the correct transfer decision is obtained by executing a family-specific target-world verifier rather than by an annotator or by the generator's perturbation identity. Four heterogeneous exact-verifier families were frozen before confirmatory outcome access: mapped directed-flow feasibility, Horn-style logical entailment, dimensional/unit transformations and finite deterministic state-transition systems. The first development instrument failed the preregistered coordinate-ablated-twin circularity attack and was preserved as negative history rather than promoted. A redesigned development instrument passed the registered checks; its paired variance gave a required decidable sample of 431 for the registered 0.05 MDE, and seed `2026081202` with $n = 576$ total cases (256 verifier-licensed, 256 verifier-rejected, 64 `CANNOT_CHECK`) was frozen and merged publicly before execution.

The registered outcome: a lexical/Jaccard control sat at chance on the 512 decidable cases (accuracy 0.496). A derived-effect ("mechanism-only") rule reached exact three-way accuracy 0.875 and accepted every valid transfer, but licensed 25% of verifier-invalid transfers; a relation/invariant-only rule reached 0.819 with 37.5% invalid false-accepts at full valid retention. The full applicability contract matched all 576 verifier decisions—every valid transfer retained, no invalid transfer licensed, every verifier-unknown case abstained. The preregistered primary statistic, paired binary-Brier reduction of the full contract against the mechanism-only control, was 0.120 with item-bootstrap 95% interval $[0.0938, 0.1481]$; the four-family sign test is $p = 0.125$, which is why a six-family extension was frozen as a separate coordinate. That extension was later executed, passed, and audited—Section 5.2.

6.2 What the battery leaves standing

The battery findings on the six-family sibling packet apply to this lane's architecture, and the manuscript must say so plainly rather than by understatement. Candidate inputs in the objective lane expose machine-readable target-world facts from which the deterministic coordinates are computed; probe G, executed on the six-family packet built from the same architecture, shows the stronger fact that the benchmark surface carries *zero* extraction signal—not merely insufficient signal—because no decision arm reads the text at all. The full arm implements the same coordinate algebra the verifier executes, so its exact agreement is a consistency property of the construction, and the control-arm deficits quantify which strata the generator contains (probe F: in the six-family packet, two thirds of the headline gain came from strata the control could not see by design).

What stands, therefore, is a *specification demonstration*: the contract's six coordinates are individually enforceable, jointly sufficient on the registered families, and the deterministic pipeline—generation, freezing, hashing, execution, paired inference—runs to completion and reproduces its registered outputs byte-for-byte. The reading "checking a subset of the six independent conditions misses violations of the others" is arithmetic once probe G is known, and we report it as arithmetic, not as an empirical transfer finding. Per the battery's probe-C corollary, no false-accept figure above is reported without its paired retention figure. The lane establishes no claim that a system recovers applicability coordinates from unstructured input, no model-level claim, and no natural-domain claim.

6.3 A demoted external-model comparator

One measured external-model result is retained, at deliberately demoted authority. An independently frozen comparator drove one fixed frontier model through the fail-closed applicability contract on $n \approx 504$ fresh exact-verifier tasks (seed `20260812`; design, primary metric, hypothesis direction and stopping rule frozen before any output was scored), against a plain DIRECT judgement and a compute-matched FREE-COT control, with the same three-way label space in every arm.

Figure 3 shows the obligation scaffold roughly halving the odds of licensing a verifier-invalid transfer while the compute-matched control moves nothing: the reduction is attributable to the obligation *structure*, not to additional reasoning tokens. Three demotions bound this result. It is one model, one seed and one family-set. It measures a structured-inference scaffold, not a smarter model, and does not help on the hardest semantic near-misses. And—decisive after the battery—the underlying benchmark's coordinates travel in pre-parsed form, so the comparator measures whether the model *applies* the named obligations to structured facts, not whether it can recover them. A cross-model, cross-seed replication on a battery-compliant benchmark is the named next coordinate before any external-model claim.

7 The natural-domain lane: preserved negative history and the open coordinate

Everything in this section is either preserved negative history or an explicitly open coordinate. No natural-domain result is claimed.

7.1 Design vocabulary

The natural-domain benchmark independently manipulates surface-semantic similarity and structural similarity, giving four cells: Q1 (high/high), Q2 (low semantic, high structural), Q3 (high semantic, low structural) and Q4 (low/low). Q2 and Q3 are load-bearing: a useful structural gate should license Q2 transfers, where vocabulary is deliberately distant but the

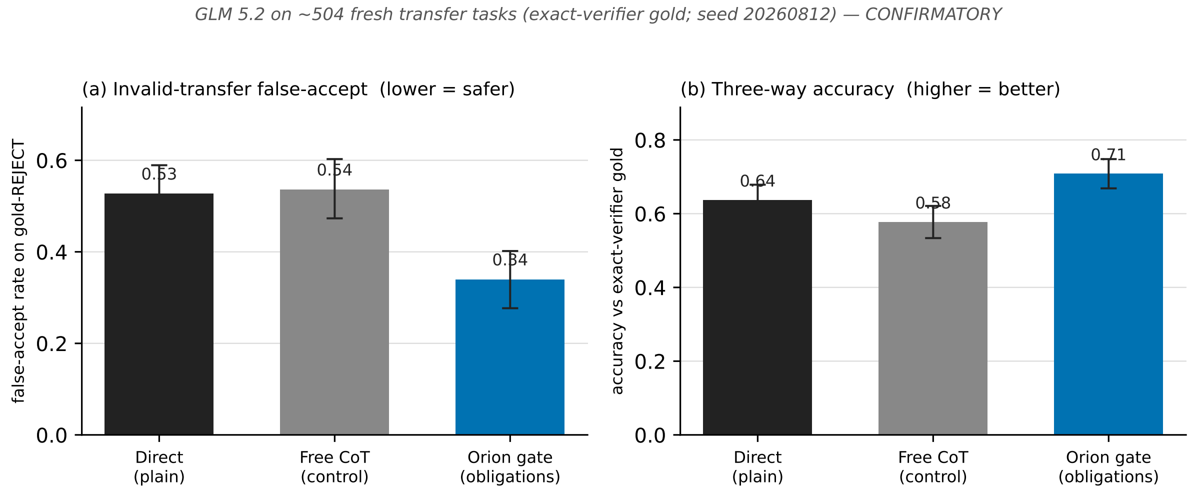

Figure 3: Demoted external-model applicability-gate comparator (one fixed frontier model; $n \approx 504$ fresh exact-verifier tasks; bootstrap 95% CIs, 20,000 resamples). **(a)** Driving the model through the fail-closed applicability gate cuts the invalid-transfer false-accept rate from $0.527\,[0.460, 0.589]$ (Direct) to $0.339\,[0.277, 0.402]$ (gate); the compute-matched Free-CoT control $(0.536\,[0.473, 0.603])$ is statistically indistinguishable from Direct. **(b)** Three-way accuracy against exact-verifier gold rises in step $(0.637 \rightarrow 0.708)$. The benchmark surface was subsequently shown to carry no extraction requirement (Section 5.2, probe G), so this measures compliance with the obligation structure over structured inputs—a scaffold effect—not prose reading or model reasoning quality; on the hardest semantic near-miss decoys the gate does not beat Direct, a real and reported weakness.

load-bearing relation/invariant structure is preserved, and reject Q3 decoys, where embedding similarity is high but the structure or boundary that justifies transfer is broken. A deterministic six-case conformance receipt (one Q2 and one Q3 case in each of three mechanism families: queue stability, positive feedback, threshold cascade) shows the implementation expresses the intended distinction—the semantic-only rule is wrong on all six hard cases by construction, the witness gate correct on all six. This is a software-conformance result, not a statistical generalization result, and provides no estimate of performance on naturally occurring mappings.

7.2 Preserved negative history

The 44-pair internal diagnostic is permanently non-confirmatory. Before inspecting predictor outcomes, a balanced Q1–Q4 proposal set, leave-one-family-out split, predictor family, metrics and stopping thresholds were frozen at Git subject `f2701f732f83`: 44 constructed proposal pairs across 11 proposed mechanism families, every coordinate proposed in the same assistant session that assembled the benchmark, with no foundation-model judgement. The deterministic pilot crossed its frozen incremental-signal thresholds (witnessed structure raised leave-one-family-out ROC-AUC from 0.914 to 1.000 and average precision from 0.903 to 1.000 over the strongest frozen lexical/tag control), but the independent-annotation gate has authority over the signal gate, and its exact verdict is `FAIL_CLOSED_MISSING_INDEPENDENT_ANNOTATION`: zero of 44 items confirmatory-eligible, no training or inference run launched. A subsequent chronology audit found the stricter cut—labels were visible during construction—so the v1 set is *permanently* an internal constructed diagnostic; later annotations cannot retroactively convert it, and the receipt is retained unchanged as negative history.

The v2.1 packet lineage, including its own repairs. The v2 lane requires a fresh label-blind item set frozen before labels or results, two genuinely independent annotations per item, distinct post-freeze adjudication and an external provenance audit. Its history is preserved with its defects: a pre-annotation identifier-leakage audit found four curator-only source identifiers containing the phrase `near_miss`, so that source set was preserved as negative history and replaced by an otherwise byte-equivalent v2.1 set with neutral identifiers; a hostile usability check found the response schemas forced `cannot_assess=false` against the frozen rubric, and the repaired schemas permit abstention only as a fully null judgement, with the compiler preserving such submissions while failing confirmatory import. The 16-item v2.1 packet was frozen with opaque identifiers, a coordinator-only linkage held outside Git, and zero external judgements at freeze.

The strong semantic control is frozen, label-blind. Before any external judgement, a content-bound control was frozen on the exact revision `953dc6f6f85a1b2d` of `BAAI/bge-reranker-v2-m3`, consuming deterministic renderings that exclude candidate invariant/boundary proposals, annotations, quadrants and outcomes. An earlier clean-environment attempt returned `CANNOT_CHECK_MODEL_ASSET_MISSING` and is preserved; label-blind descriptor jobs 3476527–3476529 subsequently reached `HARVEST_DESCRIPTOR_READY` under the preserved pre-annotation chronology, with training unauthorized throughout.

The demoted operator path is not independent confirmation. Issue #217 was closed through an explicitly demoted `AI_OPERATOR/NON_INDEPENDENT` annotation path. The first demoted train/inference pair, jobs 3476750/3476751, failed on a missing `torch` environment and remains negative history; after hardening, demoted jobs 3476753 (training) and 3476754 (inference) passed under `DEMOTED_AI_OPERATOR` authority. They establish that the demoted execution path can run; they do not attest independent external annotators, an independent adjudicator or an independent provenance auditor, and they grant no confirmatory authority. The independent-human coordinate was frozen in issue #332 as `CANNOT_OBTAIN_INDEPENDENT_EXTERNAL_HUMANS` with verdict `DEMOTED_AI_OPERATOR_PATH_COMPLETE__INDEPENDENT_REVIEW_ABSENT`; the residual continues under open successor #359. The correct scientific status is `INDEPENDENT_CONFIRMATION_ABSENT`.

Separately, the prose-extraction instrument is not probative. The natural-domain question has an extraction half—can the contract's coordinates be recovered from prose at all?—and the current instrument for it terminated at `INSTRUMENT_NOT_PROBATIVE__TEMPLATE_INVERSION` (Section 5.4). Prose-extraction natural-domain items currently sit at $n = 16$ with internally authored coordinates.

7.3 The open coordinate, stated precisely

Before any external judgement, a label-blind power study registered a primary paired Brier-reduction MDE of 0.05 and simulated paired bootstrap/sign-flip resolution under plausible item-level noise. The frozen 16-item packet is *underpowered* for that MDE: adequate resolution requires roughly $n \approx 48$ items across the registered noise envelope. The lane is therefore scoped `CONFIRMATORY_PACKET_POWER_LIMITED`: indistinguishable outcomes are inconclusive rather than negative evidence, confirmatory refutation claims remain blocked when intervals include zero, the power limitation is not repaired by the demoted path, and the item set cannot be enlarged post-label and represented as the same confirmatory epoch.

The binding requirement is an externally labelled natural-domain packet at $n \approx 48$ whose label authors are independent of the instrument's author. Two routes are admissible, and both are governed by the same admission gate (Section 8): a solicited independent-annotation packet under the open successor issue #359 (two genuinely external annotators, a distinct adjudicator, a distinct provenance auditor); or a published third-party-labelled corpus whose items fit the transfer-instrument shape, where author-independence holds mechanically because the labels were authored before and outside this programme.

7.4 The corpus route, executed: a measured negative

The second route has now been executed once, end to end, under a protocol frozen before first dataset contact (`research/paper2_external_corpus_v1/`). A bounded search selected ARN [25]: 1,095 query–choice–choice triples over the four quadrants of near/far surface similarity against analogy/distractor status, with gold system mappings deriving from shared proverbs in the crowdworked ePiC corpus and triples vetted by the ARN authors—labels whose authors are mechanically independent of this programme. The registered deterministic reducer was admitted by the implemented admission gate at external-label kind (scramble-sensitive; obstruction-surfacing on the known-answer calibration source), the schema was bound by a committed amendment before any arm scoring, and the falsifiability battery of Section 5.1 passed in full: text destruction changed the witness arm's output on 1,542 of 1,542 confirmatory pairs and collapsed its accuracy to the shuffled-gold null; the shuffled-gold negative control failed the advantage gate as required; no trivial arm attained the joint property; both compared arms carried per-item loss variance.

Because the battery passed, the outcome is a measurement, and the outcome is negative: `NEGATIVE__CAPABILITY_ABSENT`. On 1,542 held-out pairs (split by proverb so no system mapping crosses the development/confirmatory boundary), the witness arm's paired-Brier advantage over the strongest frozen non-structural control was -0.016 with item-bootstrap 95% interval $[-0.0498, +0.0162]$—the upper bound below the registered 0.05 MDE—and its joint property failed with valid-transfer retention 0.023 at invalid false-accept 0.044 (reported as a pair, per probe C). On the Q2-analogue quadrant, far analogies, the gate retained 1.3% ($n = 386$). The mechanism is exactly the extraction gap this paper names: cross-domain analogies share almost no surface vocabulary, and an admissible deterministic reducer sees only surface roles, so the fail-closed gate collapses to near-total rejection at chance-level Brier.

This result relocates the open coordinate rather than discharging it. The residual is *not* label availability: third-party labels exist, were acquired, and were used at a sample size ($n = 1{,}542 \gg 48$; CI half-width ± 0.033) that leaves no power excuse. The residual is a measured extraction-capability gap: no admissible reducer in this programme recovers system-level structure from natural narratives. The scope of the negative is the registered reducer and this programme, not all possible reducers; the named successor is a capable learned extractor, which owes the same admission gate plus a contamination declaration (ARN has been public since 2023). Until such a reducer exists and passes, the generality question this paper inherits from its earlier drafts remains open—now with the blocking capability measured rather than asserted—and the solicited-annotation route under #359 remains open for the frozen v2.1 lane.

8 The open programme: mandatory comparators, admission, and stopping rules

This section is a preregistered plan, not a result. It exists so that the open coordinate of Section 7.3 cannot later be discharged against a weakened comparator set or an instrument that fails the battery.

8.1 Mandatory comparator arms

The historical packets compared the full contract against lexical, relational, mechanism-only and coordinate-ablated arms—projections of itself. Given the parent analysis of Section 3, no novelty-bearing empirical claim may be made until the following arms are present.

1. **Transportability oracle.** On any benchmark item expressible as a causal transport problem, run *sID* (or an equivalent complete transport decision) with the selection diagram supplied [3, 5]. Wherever the oracle is defined and agrees with the contract, the contract's residual is confined to the cases the oracle cannot express—which is exactly the claim that should then be made. This is the single most important missing comparator.

2. **Structure-mapping baseline.** SME-style structural evaluation over the role/relation content [10, 11], thresholded to a decision, to establish whether the mapping conjunct carries any measured advantage.

3. **Matched-coverage selective prediction.** A confidence-thresholded abstainer and a conformal abstainer under shift [13, 15], both tuned to match the contract's abstention *rate*. This is the fair test of whether structure-triggered `CANNOT_CHECK` beats threshold-triggered abstention at equal coverage.

4. **Mechanism-alignment LLM judge under matched context.** Same information, same token budget, asked directly to judge transfer validity. The retained comparator of Section 6.3 is one model, one seed, one family-set, on a pre-battery benchmark.

5. **Trivial gates.** `always_reject`, `always_accept`, `always_cannot_check`—established as necessary by probe C: any false-accept figure reported without its paired valid-transfer retention figure is uninterpretable.

Every future instrument must satisfy the design requirements of Section 5.5 before its outcomes are read; a benchmark that fails the battery is repaired in a new versioned epoch, never rescored.

8.2 The admission gate for reducers

The extraction half of the open coordinate is governed by an implemented admission gate (`src/rakl/reduction_validation.py`) whose checks are each traceable to a measured failure of this programme: a reducer returning identical structure from scrambled text is rejected outright—one scramble-invariant source is disqualifying, with no tunable threshold (the probe-G failure shape); a reducer that fails to surface the known obstruction in a fixed calibration source is rejected (obstruction-blind distillation is unsound for navigation); and a reducer whose validation labels are absent or share the reducer's author is capped at an asserted-authority floor (the template-inversion finding: self-authored validation measures the authorship). Admission above the floor requires external labels whose author is distinct from the reducer's author. Published third-party-labelled corpora satisfy this author-independence mechanically; this route has been exercised once, end to end, with a fully passing battery and an honest negative terminal (Section 7.4). A successor reducer enters through the same gate—with a contamination declaration when it embeds trained components—and a bounded search that finds no suitable corpus is itself a reportable terminal, never a licence to lower the gate.

8.3 Stopping rules

The programme stops or narrows if: the structural score adds no transfer information beyond the strongest calibrated semantic control [24]; Q3 semantic decoys are frequently licensed; structure labels are unstable across independent annotators or extraction methods; a parent formalism or system—transportability where expressible, SME, selective prediction at matched coverage, or a SKILL.nb-style gate [18]—matches the contract's decisions; or the instrument for a claim fails the battery and no battery-compliant successor exists. A null or indistinguishable outcome on an underpowered packet is inconclusive, never negative evidence; a failed instrument is preserved and versioned, never rescored.

Earlier drafts of this manuscript carried training-data-selection, cost-amortization and learner-conditioned-curriculum hypotheses with their own parent sets. Those are withdrawn from this paper's claims entirely: they are not established, not refuted, and not part of the present contribution; their preregistered plans remain in the repository's research archive and would require their own instruments, comparators and battery compliance before any future use.

Code, materials and AI-use disclosure

A public research artifact accompanies this paper. Frozen protocols, receipts, audit batteries and machine-readable negative results are retained alongside the manuscript so that publication prose cannot silently replace missing evidence; in particular, every instrument refutation reported here (the non-falsifiable six-family gate, the serialization-fidelity reading of the controlled-witness instrument, and the template-inversion terminal of the prose instrument) is preserved as a first-class artifact rather than summarized away. Language models were used as research, coding and drafting tools; they are not authors. Same-context analysis, including the audit batteries, is explicitly distinguished from independent review, and no external judgement is represented as completed until a hash-bound response, adjudication and provenance audit exist. Author identity, affiliation and corresponding email are given on the title page. Funding, competing-interest, acknowledgement and licensing declarations remain author-supplied submission metadata and are not inferred by the repository.

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